Thanh-Khoi Nguyen

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PROFILE

My research focuses on **3D scene understanding and spatial reasoning**, particularly geometry-grounded vision-language methods using **3D Gaussian Splatting**. I am interested in extending these approaches toward dynamic scene understanding and spatially grounded world models.

EDUCATION

University of Science, VNU-HCM (HCMUS) Sep 2023 – May 2027
B.Sc. in Computer Science; Honor Programme – Information Technology Talent Program

- GPA: 3.94/4.0 – **Top 2%** of faculty; Dean's Honor List 2024 (Top 3%), 2025 (Top 2%)
- Admitted to the IT Talent Program (**top 45 students** selected faculty-wide by competitive examination)
- Selected coursework (10-point scale): Computer Vision (9.30), Fundamentals of Artificial Intelligence (9.70), Linear Algebra (10.00), Probability and Statistics (10.00), Computer Systems (10.00), Data Structures and Algorithms (9.40)

RESEARCH AND PROFESSIONAL EXPERIENCE

Research Engineer Jul 2026 – Oct 2026
Neoviews | United Kingdom (Remote)

- Develop 3D human pose and shape estimation methods from monocular sports footage for real-world deployment.
- Work on single-view 3D reconstruction pipelines, addressing challenges in occlusion, fast motion, and camera viewpoint variation specific to sports scenarios.

Research Student Mar 2026 – Sep 2026
Saigon AI Hub (SAIH), Applied AI Lab - VNG Group & VNU-HCM | Ho Chi Minh City, Vietnam

- One of 10 inaugural research groups at Vietnam's first open AI hub, selected for Google Labs' AI Futures Fund.
- Study visual, temporal, and language grounding in sports video for structured scene reasoning and spatial analysis.

Undergraduate Researcher Jun 2024 – Present
Software Engineering Lab (SELab), VNU-HCMUS | Advisor: Assoc. Prof. Minh Triet Tran

- Research **3D/4D Gaussian representations** for view-dependent referring segmentation, online semantic mapping, and object-part retrieval, targeting spatial understanding of complex scenes.
- Develop and evaluate methods for geometry-grounded semantic reasoning, including training-free 3DGS segmentation and view-conditioned spatial grounding.
- Co-authored **11 accepted or published papers** across 3D vision and multimodal understanding, including work at SIGGRAPH Asia 2026, MMM 2027, and *Computers & Graphics*; two additional 3DGS paper is under review.

PUBLICATIONS

[1] **VISOR: View-Conditioned Spatial Reasoning for 3D Gaussian Referring Segmentation**
Thanh-Khoi Nguyen, Minh-Triet Tran*
SIGGRAPH Asia 2026, Poster Track (CORE A)* Accepted

[2] **DSSR-3D: Decoupled Reasoning for View-Dependent Referring in 3D Gaussians**
Thanh-Khoi Nguyen, Thien-Phuc Tran, Minh-Triet Tran*
MMM 2027 (CORE B) Accepted

[3] **RecallGS: Geometry-Grounded Recall for Online Semantic Gaussian Splatting**
Thanh-Khoi Nguyen, Thien-Phuc Tran, Minh-Triet Tran*
SOICT 2026 Accepted

[4] **Query Rewriting for Complex Object Segmentation in 4D Gaussian Representations**
Thanh-Khoi Nguyen, Thien-Phuc Tran, Minh-Triet Tran*
SOICT 2026 Accepted

[5] **Action-Aligned Retrieval with Pairwise Multimodal Reranking for Text-Based Person Anomaly Search**
Thanh-Khoi Nguyen†, Thanh-Nhan Vo†, Trong-Thuan Nguyen, Minh-Triet Tran*
AI City Workshop @ ECCV 2026 (CORE A)* Accepted

[6] **A Top-Down Framework for Metric-Scale Athlete Localization from Single Broadcast Frames**
Thanh-Khoi Nguyen, Hoang-Phuc Nguyen, Linh-Huynh, Minh-Triet Tran*
Proc. ICMV 2026 (CORE C) Accepted

[7] **TreeSoc: Tree-Structured Dynamic Reasoning and Tool Synergy for Soccer Video Understanding**
Thanh-Nhan Vo, **Thanh-Khoi Nguyen**, Trong-Thuan Nguyen, Trung-Hoang Le, Minh-Triet Tran*
Proc. ICMV 2026 (CORE C) Accepted

[8] **SnapSeek 4.0: Bridging Temporal Dynamics and Regional Semantics for Interactive Lifelog Retrieval**
Minh-Quan Ho-Le, **Thanh-Khoi Nguyen**, Van-Tu Ninh, Cathal Gurrin, Minh-Triet Tran*
ACM LSC Workshop @ ICMR 2026 (ICMR: CORE B) Published

[9] **From Text to Thumbnail: A Unified Framework for Automated News Image Generation and Evaluation**
Thanh-Khoi Nguyen, Hai-Dang Nguyen, Minh-Triet Tran*
Proc. SOICT 2025 Published

[10] **Less is More: A Dual-Filtering Perspective on Prompt Design for News Thumbnail Generation**
Thanh-Khoi Nguyen, Hai-Dang Nguyen, Minh-Triet Tran*
CEUR Workshop Proceedings, MediaEval 2025 Published

[11] **SHREC GS-3DORC: Towards the Advancements of 3D Gaussian Splatting Object Part Retrieval**
Thien-Phuc Tran, Minh-Quang Nguyen, **Thanh-Khoi Nguyen**, Nam-Quan Nguyen, Minh-Triet Tran*
Computers & Graphics (Elsevier, Q2), Vol. 132, 2025 DOI

* Corresponding author † Equal contribution

Manuscripts Under Review

[1] **ROAM: Routing Observation Alternatives from Mask Agreement for Training-Free 3DGS Segmentation**
Thanh-Khoi Nguyen, Minh-Triet Tran
Submitted to ICAART 2027 Under review

[2] **Training-Free Adaptive Grouping for Open-Vocabulary 3D Gaussian Segmentation**
Thanh-Khoi Nguyen, Minh-Triet Tran
Submitted to ACIIDS 2027 Under review

SELECTED HONORS AND AWARDS

- **1st Place, SoccerNet 2026 Spiideo SynLoc Challenge** 2026
  International sports AI benchmark for single-frame world-coordinate athlete localization; 97.67 mAP, surpassing baseline by over 21%.
- **5th Place, SoccerNet 2026 VQA Track** 2026
  Developed an agent-based system for structured visual question answering over soccer broadcast video.
- **Top 3, ACM MM 2026 TRIDENT Grand Challenge (Video Track)** 2026
  Tri-modal explainable deepfake forensics benchmark at ACM Multimedia (CORE A*); method GUARD achieved TCS 51.08 (detection 65.94, hallucination robustness 37.20, perception 45.13).
- **Best Paper Award, Youth Science Conference (YSC2026)** 2026
  Selected from all accepted submissions at the HCMUS university-level research conference.
- **Odon-Vallet Scholarship** 2023, 2026
  Nationally selective scholarship for exceptional academic merit in Vietnam.
- **Dean's Honor List** 2024 (Top 3%), 2025 (Top 2%)
  Highest-performing students in the faculty; two consecutive years.
- **Merit-Based Scholarship for Outstanding Students** 2023, 2024
  Awarded to top 5% of students faculty-wide; two consecutive years.

SKILLS

3D Vision & AI: 3D Gaussian Splatting, 3D/4D Segmentation, 3D Scene Understanding, Spatial Reasoning, Computer Vision, Machine Learning
Programming Languages: C/C++, Python, Java, Kotlin, SQL, R
Frameworks/Tools: PyTorch, Open3D, TensorFlow, Keras, Hugging Face, Docker, Git, Android Studio, MySQL, PostgreSQL
Languages: Vietnamese (native speaker), English (professional working proficiency, IELTS 7.0)